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Can Tobacco Control Policies Prevent Tooth Loss? A Difference-in-Differences Study

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- Smoking is the dominant modifiable cause of periodontitis and tooth loss
- Cigarette taxes, prices and smoke-free laws reduce smoking
- Tobacco policy is rarely evaluated against oral-health outcomes

Primary aim (substantive)

Estimate the effect of U.S. state tobacco-control policies on smoking and on tooth loss

Secondary aim (methodological)

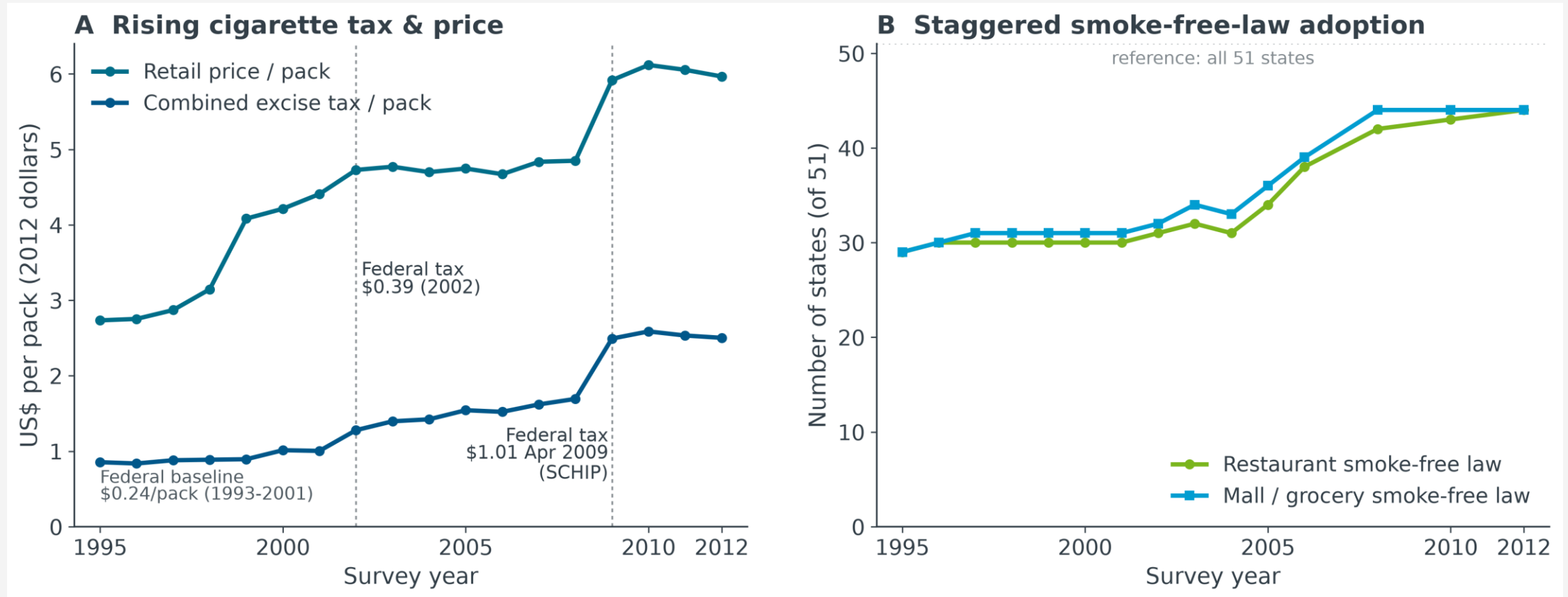
Benchmark heterogeneity-robust DiD estimators against two-way fixed effects in a design combining staggered binary and continuous treatments

	Operationalisation
Sample	Behavioral Risk Factor Surveillance System (BRFSS), 1995–2012; 15 cycles, N = 3,840,519, 51 states
Treatments	Cigarette tax and price (\$/pack, continuous); restaurant and mall/grocery smoke-free laws (staggered)
Outcomes	Complete edentulism; loss of ≥ 6 teeth
Controls	Positive: current smoking, quitting; negative: seat-belt use, smoke-detector

Estimand: the average treatment effect on the treated (ATT), under parallel trends in the mean untreated potential outcome

U.S. state tobacco-control policies, 1995–2012

A rising cigarette tax and price (left) and smoke-free-law adoption (right) — the variation the difference-in-differences design exploits



Problems of two-way fixed effects: staggered and continuous treatments

- Staggered adoption: already-treated units serve as controls — forbidden comparisons
- Continuous dose: no stayers — the 1996 federal tax moved every state off baseline

$$\hat{\beta}^{\text{TWFE}} = \sum_k s_k \hat{\text{DD}}_k, \quad s_k \geq 0, \quad \sum_k s_k = 1$$

Goodman-Bacon (2021): TWFE is a weighted average of all observed 2x2 DiDs; these weights are non-negative, but some of the 2x2 comparisons are contaminated

$$\beta^{\text{TWFE}} = \sum_{g,t} w_{g,t} \text{ATT}(g, t), \quad \exists w_{g,t} < 0$$

de Chaisemartin–D'Haultfœuille (2020): rewritten over group-time effects, the weights need not be non-negative — the sign of the estimand is not guaranteed

$$\text{ATT}(d_j) - \text{ATT}(d_{j-1}) = \text{ACRT}(d_j) + \text{ATT}(d_{j-1} \mid d_j) - \text{ATT}(d_{j-1})$$

With a continuous dose, comparing dose levels returns the ^{causal} response (ACRT) plus a ^{selection on dose} selection term, which vanishes only under strong parallel trends (Callaway et al. 2024)

- **Group-time ATT** — estimate each cohort's effect separately, using only never- or not-yet-treated controls (Callaway–Sant'Anna 2021; Sun–Abraham 2021)
- **Imputation** — fit the untreated model on clean cells, predict $Y(0)$ for treated cells, average the residuals (Borusyak–Jaravel–Spiess 2024; Gardner two-stage 2022)
- **Extended TWFE** — saturate the regression with cohort $\times$ period terms so the implied weights stay non-negative (Wooldridge 2025)
- Continuous dose handled as high-dose cohorts: tax $\geq$ \$3, price $\geq$ \$6 per pack

$$ATT(g, t) = E[Y_t - Y_{g-1} \mid G_g = 1] - E[Y_t - Y_{g-1} \mid D_t = 0]$$

The change from the last untreated period $g-1$ to period t in cohort g , minus the same change in a clean comparison group; aggregated with weights chosen by the analyst

No stayers is demonstrated, not assumed: all states first change treatment at the same date, so Design Restriction 1 fails (de Chaisemartin et al. 2024)

Results: cigarette tax, smoking, and tooth loss

Smoking — percentage points (95% CI) per +\$1.00/pack

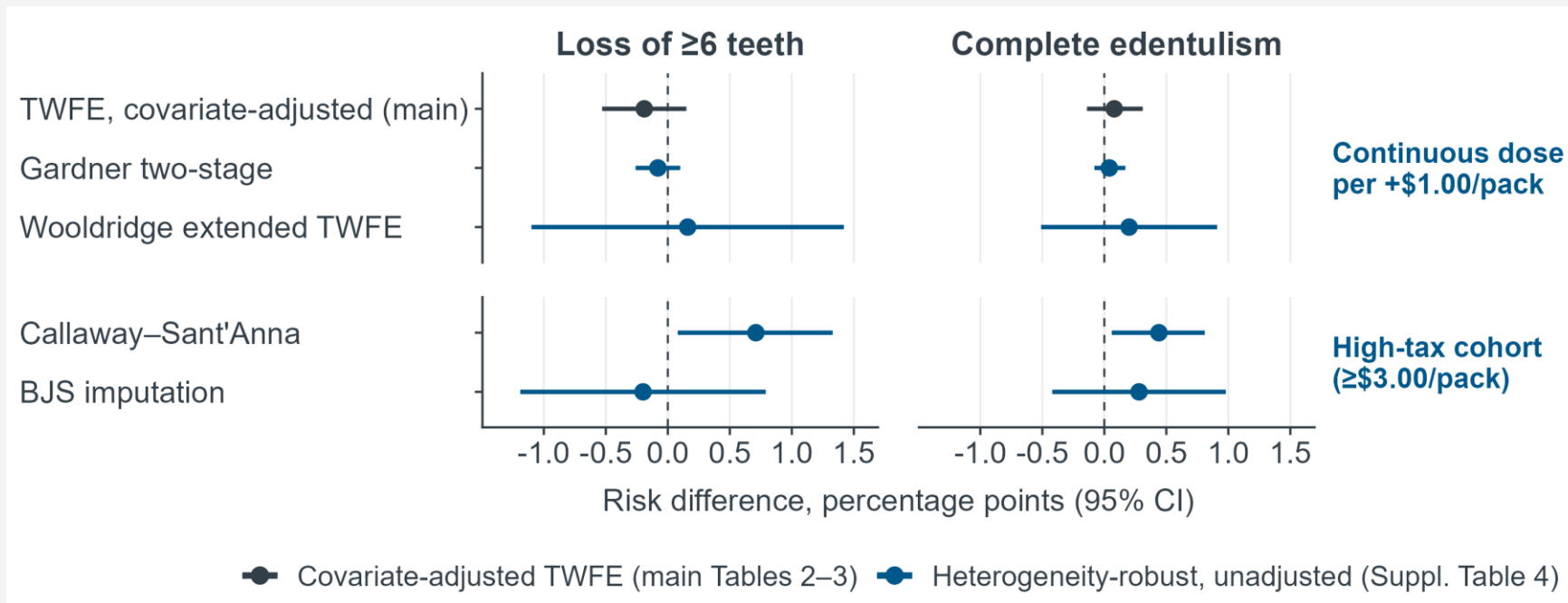
Outcome	Tax, current year	Tax, 10-year average
Current smoking	−0.35 (−0.56, −0.13)	−0.47 (−0.95, 0.01)
Quitting	0.51 (0.19, 0.82)	0.07 (−0.53, 0.67)

Tooth loss — percentage points (95% CI) per +\$1.00/pack

Outcome	Tax, current year	Tax, 10-year average
Complete edentulism	0.08 (−0.14, 0.31)	0.39 (−0.12, 0.89)
Loss of ≥6 teeth	−0.19 (−0.53, 0.15)	−0.44 (−1.17, 0.30)

Covariate-adjusted two-way (state and year) fixed-effects estimates, linear probability model, unweighted individual-level data; standard errors clustered by state. Retail price shows similar estimates.

Estimator comparison: where the methods agree and diverge



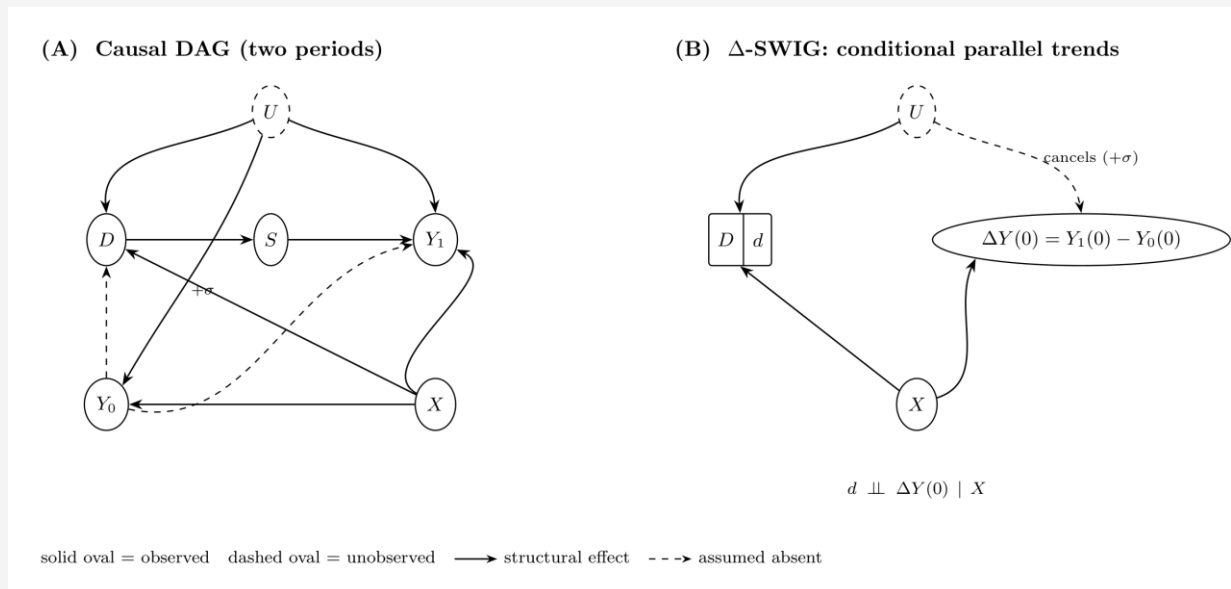
Why the estimates differ

- Different estimands: continuous per +\$1.00 dose versus binary $\geq \$3.00$ -cohort ATT
- Robust estimators unadjusted on a weighted state-year panel; main models covariate-adjusted
- Callaway–Sant'Anna rests on small, late, biennially observed cohorts; imputation, more stable, stays null

Sun–Abraham omitted: variance undefined and the group-time estimator does not converge for these outcomes

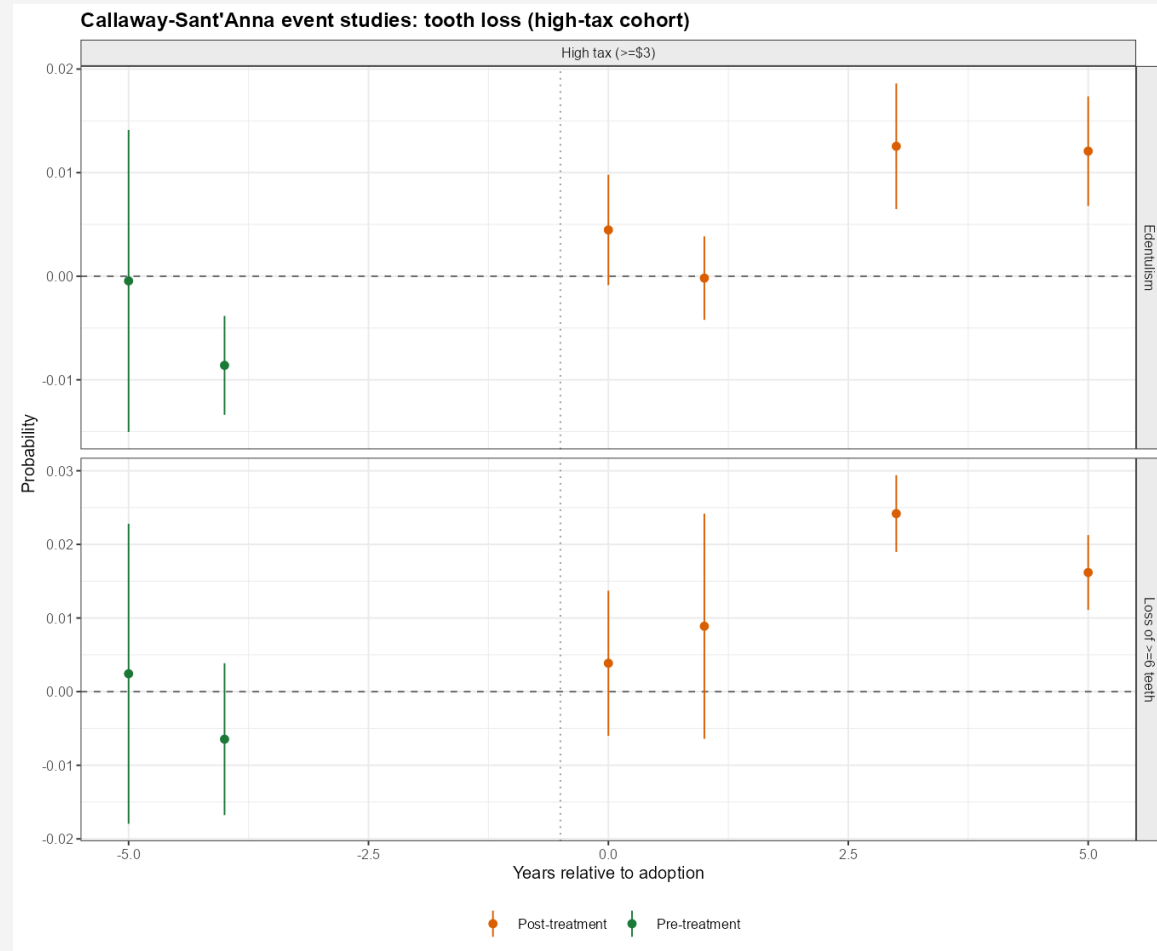
$$E[\Delta Y(0) \mid D = 1] = E[\Delta Y(0) \mid D = 0]$$

- Untestable: pre-trends neither necessary nor sufficient (Roth 2022); judge magnitude
- The causal graph locates the assumption; it does not establish it (Renson et al. 2026; Knaus & Pfleiderer 2026)
- Rambachan–Roth relative-magnitude bound ($M = 2$): CI widens to $[-9.0, 11.1]$ pp



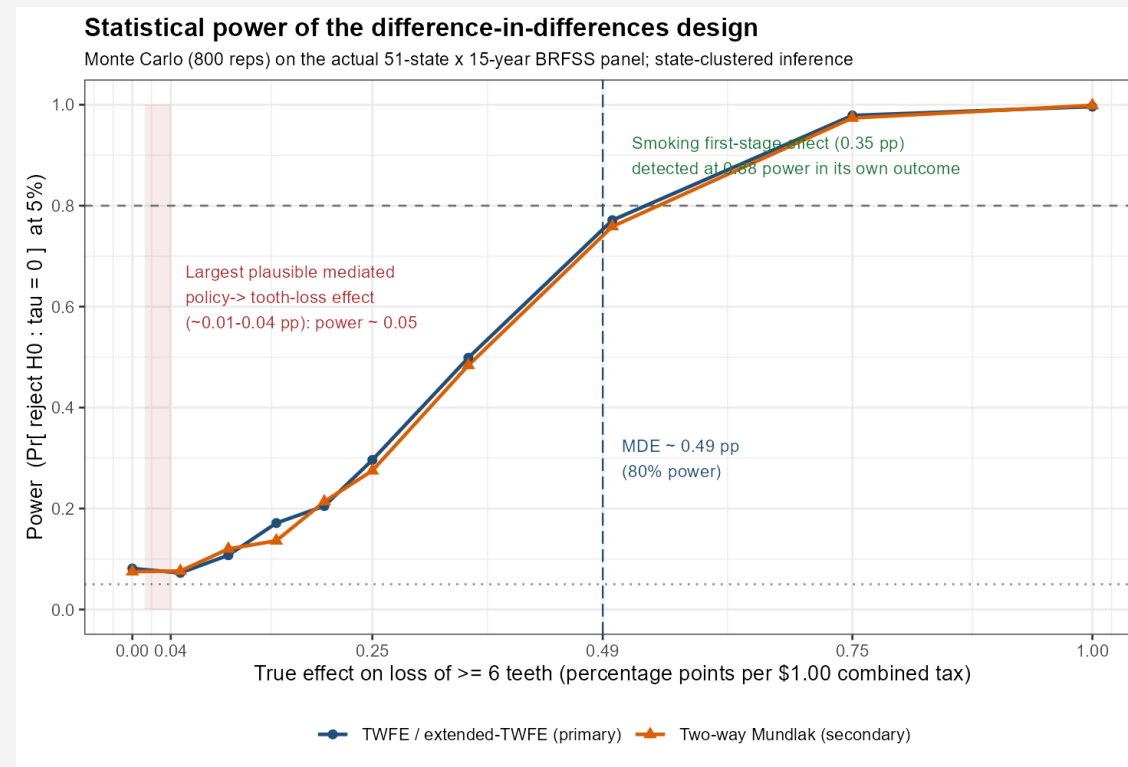
Callaway–Sant'Anna event study (Roth 2026)

Universal base period: pre-treatment coefficients read as pre-trends — flat before adoption; post-treatment coefficients are non-negative, i.e. no protective dynamic effect



Is the null informative? Minimum detectable effect and plausibility

- Minimum detectable effect 0.49 pp; the plausible mediated effect is only 0.01–0.04 pp
- At that size power is ≈ 0.05 ; the ceiling is set by between-state variation and 51 clusters (Bertrand et al. 2004)



A design limit, not evidence of absence — an equivalence statement (Rainey 2014; Lakens et al. 2018)

1. Cigarette taxes and prices reduce smoking; no detectable effect on tooth loss over 1995–2012
2. The null is a limit of the design: the plausible mediated effect lies far below the minimum detectable effect
3. Estimators agree on the positive control; for tooth loss they diverge under sparse, late high-dose cohorts
4. Continuous-dose DiD is the binding constraint: no stayers, and dose contrasts carry selection on dose



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Thank you

Paper in print in the *Journal of Dental Research*

Slides and analysis code:

github.com/BaumeisterS/DGEPI2026_tobacco_policy_tooth_loss

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